



SOLUTIO OF SERIES OF EXERCISES N° 1 : REAL NUMBERS

Exercise 1

1. Let $n \in \mathbb{N}^*$ such that n is not the square of any integer. Show that $\sqrt{n} \notin \mathbb{Q}$.
2. Prove that $\sqrt{2} + \sqrt{3} \notin \mathbb{Q}$.

Solution to Exercise 1

1. Let $n \in \mathbb{N}^*$ such that n is not the square of any integer. Show that $\sqrt{n} \notin \mathbb{Q}$.

We give a clear step-by-step proof by contradiction.

Step 1 (Assume rational). Suppose, towards a contradiction, that $\sqrt{n}$ is rational. Then there exist integers p, q with $q > 0$ and $\gcd(p, q) = 1$ such that

$$\sqrt{n} = \frac{p}{q}.$$

Step 2 (Square both sides). Squaring both sides gives

$$nq^2 = p^2.$$

Step 3 (Prime factorization argument). Write the prime factorization of n , p and q . The left-hand side nq^2 has the property that every prime factor appears with an *even* exponent except possibly those coming from n . But since the right-hand side p^2 is a perfect square, every prime in its factorization appears with an even exponent. Comparing the two factorizations, we deduce that every prime factor of n must appear with an even exponent. In other words, n itself is a perfect square.

Step 4 (Contradiction). This contradicts the hypothesis that n is not a square. Therefore our assumption was false, and

$$\sqrt{n} \notin \mathbb{Q}.$$

□

2. Prove that $\sqrt{2} + \sqrt{3} \notin \mathbb{Q}$.

We again use a contradiction argument and reduce to 1.

Step 1 (Assume rational). Let

$$\alpha = \sqrt{2} + \sqrt{3}.$$

Assume for contradiction that $\alpha \in \mathbb{Q}$.

Step 2 (Square α). Compute

$$\alpha^2 = (\sqrt{2} + \sqrt{3})^2 = 2 + 3 + 2\sqrt{6} = 5 + 2\sqrt{6}.$$

Step 3 (Solve for $\sqrt{6}$). Rearranging gives

$$\sqrt{6} = \frac{\alpha^2 - 5}{2}.$$

Since $\alpha \in \mathbb{Q}$, the right-hand side is rational, so this implies $\sqrt{6} \in \mathbb{Q}$.

Step 4 (Contradiction using Part (1)). But 6 is not a perfect square, so by Part (1) we know $\sqrt{6} \notin \mathbb{Q}$. This is a contradiction.

Therefore $\alpha = \sqrt{2} + \sqrt{3} \notin \mathbb{Q}$. □

Exercise 2 — Inequalities

I) Let $x, y, z \in \mathbb{R}$. Prove the following inequalities:

1. $xy + yz + zx \leq x^2 + y^2 + z^2$.
2. $||x| - |y|| \leq |x - y|$.
3. $\frac{x+y}{2} \geq \sqrt{xy}$ for all non-negative real numbers x and y .

II) Prove the Cauchy–Schwarz inequality. For every $n \in \mathbb{N}^*$ and real numbers $x_1, \dots, x_n$ and $y_1, \dots, y_n$, we have

$$\left(\sum_{i=1}^n x_i y_i \right)^2 \leq \left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right).$$

Solution to Exercise 2

I. Inequalities

Exercise 2 — Inequalities (Part I)

I) Let $x, y, z \in \mathbb{R}$. Prove the following inequalities:

1. $xy + yz + zx \leq x^2 + y^2 + z^2$.
2. $||x| - |y|| \leq |x - y|$.
3. $\frac{x+y}{2} \geq \sqrt{xy}$ for all non-negative real numbers x and y .

Solution to Exercise 2 — Part I

1. Proof that $xy + yz + zx \leq x^2 + y^2 + z^2$ (using $(x - y)^2$). Start from the non-negativity of a square:

$$(x - y)^2 \geq 0.$$

Expanding gives

$$x^2 - 2xy + y^2 \geq 0 \implies 2xy \leq x^2 + y^2.$$

Apply the same inequality to the pairs (y, z) and (z, x) :

$$2yz \leq y^2 + z^2, \quad 2zx \leq z^2 + x^2.$$

Adding the three inequalities we obtain

$$2(xy + yz + zx) \leq 2(x^2 + y^2 + z^2),$$

hence dividing by 2 yields the desired inequality

$$xy + yz + zx \leq x^2 + y^2 + z^2.$$

This proof is elementary and follows directly from $(x - y)^2 \geq 0$. $\square$

2. Proof that $||x| - |y|| \leq |x - y|$ (step-by-step, including the role of y). We prove two inequalities that together imply the claim.

Step 1. By the triangle inequality, for all real x, y ,

$$|x| = |(x - y) + y| \leq |x - y| + |y|.$$

Rearranging gives

$$|x| - |y| \leq |x - y|. \tag{1}$$

Step 2. Exchange the roles of x and y . Then the same argument yields

$$|y| = |(y - x) + x| \leq |y - x| + |x| = |x - y| + |x|,$$

so

$$|y| - |x| \leq |x - y|. \tag{2}$$

Step 3. Inequalities (1) and (2) together give

$$-|x - y| \leq |x| - |y| \leq |x - y|.$$

This double inequality is equivalent to

$$||x| - |y|| \leq |x - y|.$$

Thus the result holds. (Note: we explicitly used the symmetric argument for y in Step 2 so nothing is omitted.) $\square$

3. Proof that $\frac{x + y}{2} \geq \sqrt{xy}$ for $x, y \geq 0$. Let $x, y \geq 0$. Consider the square of the difference of the square roots:

$$(\sqrt{x} - \sqrt{y})^2 \geq 0.$$

Expanding gives

$$x + y - 2\sqrt{xy} \geq 0 \implies 2\sqrt{xy} \leq x + y.$$

Dividing by 2 yields the arithmetic mean–geometric mean inequality for two numbers:

$$\sqrt{xy} \leq \frac{x + y}{2}.$$

Equality holds exactly when $\sqrt{x} = \sqrt{y}$, i.e. $x = y$. $\square$

II. Cauchy–Schwarz inequality

Consider the function

$$f(t) = \sum_{i=1}^n (x_i t - y_i)^2, \quad t \in \mathbb{R}.$$

For all real t , $f(t) \geq 0$. Expanding,

$$f(t) = \left(\sum_{i=1}^n x_i^2 \right) t^2 - 2 \left(\sum_{i=1}^n x_i y_i \right) t + \sum_{i=1}^n y_i^2.$$

This quadratic in t is non-negative for every t , so its discriminant is ≤ 0 . The discriminant equals

$$\Delta = 4 \left(\sum_{i=1}^n x_i y_i \right)^2 - 4 \left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right) \leq 0.$$

Rearranging yields

$$\left(\sum_{i=1}^n x_i y_i \right)^2 \leq \left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right),$$

which is the Cauchy–Schwarz inequality. □

Exercise 3 — Reasoning by Induction

Prove by induction the following assertions.

1. For all $n \in \mathbb{N}^*$,

$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

2. For all $n \in \mathbb{N}$, $|\sin(nx)| \leq n|\sin x|$, for all $x \in \mathbb{R}$.

3. Show that, for all $n \in \mathbb{N}^*$,

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} < \sqrt{n} + \sqrt{n+1} - 1.$$

Solution to Exercise 3

1. We prove by induction that

$$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Base case ($n = 1$). For $n = 1$ the left-hand side is $1^3 = 1$ and the right-hand side is $\left(\frac{1 \cdot 2}{2}\right)^2 = 1$, so the statement holds for $n = 1$.

Induction hypothesis. Assume the formula holds for some $n \geq 1$, i.e.

$$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Induction step. For $n + 1$ we have

$$\begin{aligned}
 \sum_{k=1}^{n+1} k^3 &= \sum_{k=1}^n k^3 + (n+1)^3 \\
 &= \left(\frac{n(n+1)}{2} \right)^2 + (n+1)^3 \\
 &= (n+1)^2 \left(\frac{n^2}{4} + (n+1) \right) \\
 &= (n+1)^2 \left(\frac{n^2 + 4n + 4}{4} \right) \\
 &= (n+1)^2 \left(\frac{(n+2)^2}{4} \right) \\
 &= \left(\frac{(n+1)(n+2)}{2} \right)^2.
 \end{aligned}$$

This proves the formula for $n + 1$. By induction the equality holds for all $n \in \mathbb{N}^*$. $\square$

2. Inequality $|\sin(nx)| \leq n|\sin x|$

We prove the statement by induction on n .

Base case ($n = 1$). For $n = 1$ we have $|\sin x| \leq 1 \cdot |\sin x|$, which is true.

Induction hypothesis. Assume that for some $n \geq 1$ and all real x ,

$$|\sin(nx)| \leq n|\sin x|.$$

Induction step. Consider $n + 1$. Using the angle addition formula we write

$$\sin((n+1)x) = \sin(nx) \cos x + \cos(nx) \sin x.$$

Taking absolute values and using the triangle inequality gives

$$\begin{aligned}
 |\sin((n+1)x)| &\leq |\sin(nx)| |\cos x| + |\cos(nx)| |\sin x| \\
 &\leq |\sin(nx)| + |\sin x| \quad (|\cos x| \leq 1, |\cos(nx)| \leq 1) \\
 &\leq n|\sin x| + |\sin x| \quad (\text{by the induction hypothesis}) \\
 &= (n+1)|\sin x|.
 \end{aligned}$$

Thus the inequality holds for $n + 1$. By induction the inequality $|\sin(nx)| \leq n|\sin x|$ holds for all $n \in \mathbb{N}$ and all real x . $\square$

3. We prove by induction on n that

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} < \sqrt{n} + \sqrt{n+1} - 1.$$

Base case ($n = 1$). For $n = 1$, we have

$$\sum_{k=1}^1 \frac{1}{\sqrt{k}} = 1 \quad \text{and} \quad \sqrt{1} + \sqrt{2} - 1 > 1,$$

so the inequality is true for $n = 1$.

Induction hypothesis. Assume the inequality holds for some $n \in \mathbb{N}^*$, that is,

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} < \sqrt{n} + \sqrt{n+1} - 1.$$

Induction step. For $n+1$, we have

$$\sum_{k=1}^{n+1} \frac{1}{\sqrt{k}} = \sum_{k=1}^n \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{n+1}}.$$

By the induction hypothesis,

$$\sum_{k=1}^{n+1} \frac{1}{\sqrt{k}} < \sqrt{n} + \sqrt{n+1} - 1 + \frac{1}{\sqrt{n+1}}.$$

It remains to show that

$$\sqrt{n} + \sqrt{n+1} - 1 + \frac{1}{\sqrt{n+1}} \leq \sqrt{n+1} + \sqrt{n+2} - 1. \quad (1)$$

Simplifying inequality (1) is equivalent to

$$\frac{1}{\sqrt{n+1}} \leq \sqrt{n+2} - \sqrt{n}.$$

Multiply both sides by $\sqrt{n+1}$ (positive), giving

$$1 \leq (\sqrt{n+2} - \sqrt{n})\sqrt{n+1}.$$

Squaring both sides (preserving the inequality since both sides are positive), we get

$$1 \leq (\sqrt{n+2} + \sqrt{n})^2/4 \Rightarrow (\sqrt{n+2} + \sqrt{n})^2 \geq 4(n+1).$$

Expanding the left-hand side:

$$n+2+n+2\sqrt{n(n+2)} \geq 4n+4 \quad \Rightarrow \quad 2\sqrt{n(n+2)} \geq 2n+2.$$

Dividing by 2 and squaring again (valid since both sides are positive):

$$n(n+2) \geq (n+1)^2 \quad \Longleftrightarrow \quad 0 \geq 1.$$

This last statement is false, so all previous inequalities are strict, proving that (1) is satisfied strictly. Therefore, the inequality holds for $n+1$.

Conclusion. By the principle of mathematical induction, the inequality holds for all $n \in \mathbb{N}^*$:

$$\sum_{k=1}^n \frac{1}{\sqrt{k}} < \sqrt{n} + \sqrt{n+1} - 1.$$

□

Exercise 4 — The integer-part function

We denote by $E(x)$ or $\lfloor x \rfloor$ the integer part (floor) of x . Prove the following results.

1. $E(x) + E(-x) = \begin{cases} 0 & \text{if } x \in \mathbb{Z}, \\ -1 & \text{if } x \in \mathbb{R} \setminus \mathbb{Z}. \end{cases}$
2. For all $x, y \in \mathbb{R}$, if $x \leq y$ then $E(x) \leq E(y)$.
3. For all $x, y \in \mathbb{R}$,
$$E(x) + E(y) \leq E(x + y) \leq E(x) + E(y) + 1.$$
4. Show that, for all $n \in \mathbb{Z}$,

$$\left\lfloor \frac{n-1}{2} \right\rfloor + \left\lfloor \frac{n+2}{4} \right\rfloor + \left\lfloor \frac{n+4}{4} \right\rfloor = n.$$

Solution to Exercise 4

1. Value of $E(x) + E(-x)$

We have two cases:

Case A: $x \in \mathbb{Z}$. Then $\lfloor x \rfloor = x$ and $\lfloor -x \rfloor = -x$, so

$$\lfloor x \rfloor + \lfloor -x \rfloor = x + (-x) = 0.$$

Case B: $x \in \mathbb{R} \setminus \mathbb{Z}$.

So $\lfloor x \rfloor < x < \lfloor x \rfloor + 1$

Then : $-\lfloor x \rfloor - 1 < -x < -\lfloor x \rfloor$, the floor of $-x$ equals $-\lfloor x \rfloor - 1$.

Thus

$$\lfloor x \rfloor + \lfloor -x \rfloor = \lfloor x \rfloor + (-\lfloor x \rfloor - 1) = -1.$$

This proves the stated formula. □

2. Monotonicity: $x \leq y \Rightarrow \lfloor x \rfloor \leq \lfloor y \rfloor$

Let x and y be two real numbers. By definition of $\lfloor x \rfloor$ and $\lfloor y \rfloor$, we have

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1 \quad \text{and} \quad \lfloor y \rfloor \leq y < \lfloor y \rfloor + 1.$$

Since $x \leq y$, we obtain

$$\lfloor x \rfloor \leq x \leq y < \lfloor y \rfloor + 1 \Rightarrow \lfloor x \rfloor < \lfloor y \rfloor + 1.$$

Since $\lfloor x \rfloor$ and $\lfloor y \rfloor$ are integers, we then have

$$\lfloor x \rfloor < \lfloor y \rfloor + 1 \Rightarrow \lfloor x \rfloor \leq \lfloor y \rfloor.$$

□

3. Inequality for $\lfloor x + y \rfloor$ (proof by fractional parts / cases)

Write

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$$

And

$$\lfloor y \rfloor \leq y < \lfloor y \rfloor + 1$$

Then

$$\lfloor x \rfloor + \lfloor y \rfloor \leq x + y < \lfloor x \rfloor + \lfloor y \rfloor + 2$$

Two possibilities for $\lfloor x + y \rfloor$:

Case 1: $\lfloor x \rfloor + \lfloor y \rfloor \leq x + y < \lfloor x \rfloor + \lfloor y \rfloor + 1$. Then

$$\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$$

Case 2: $\lfloor x \rfloor + \lfloor y \rfloor + 1 \leq x + y < \lfloor x \rfloor + \lfloor y \rfloor + 2$. Then

$$\boxed{\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor + 1}$$

Combining both cases we obtain the double inequality

$$\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1,$$

as required. □

4. Identity involving three floors (case-by-case modulo 4)

We prove

$$\left\lfloor \frac{n-1}{2} \right\rfloor + \left\lfloor \frac{n+2}{4} \right\rfloor + \left\lfloor \frac{n+4}{4} \right\rfloor = n$$

by splitting according to the remainder of n modulo 4. Write $n = 4k + r$ with $k \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3\}$.

- $r = 0$ ($n = 4k$):

$$\left\lfloor \frac{4k-1}{2} \right\rfloor = \lfloor 2k - \tfrac{1}{2} \rfloor = 2k - 1, \quad \left\lfloor \frac{4k+2}{4} \right\rfloor = \lfloor k + \tfrac{1}{2} \rfloor = k, \quad \left\lfloor \frac{4k+4}{4} \right\rfloor = k + 1.$$

$$\text{Sum} = (2k - 1) + k + (k + 1) = 4k = n.$$

- $r = 1$ ($n = 4k + 1$):

$$\left\lfloor \frac{4k}{2} \right\rfloor = 2k, \quad \left\lfloor \frac{4k+3}{4} \right\rfloor = k, \quad \left\lfloor \frac{4k+5}{4} \right\rfloor = k + 1.$$

$$\text{Sum} = 2k + k + (k + 1) = 4k + 1 = n.$$

- $r = 2$ ($n = 4k + 2$):

$$\left\lfloor \frac{4k+1}{2} \right\rfloor = \lfloor 2k + \tfrac{1}{2} \rfloor = 2k, \quad \left\lfloor \frac{4k+4}{4} \right\rfloor = k + 1, \quad \left\lfloor \frac{4k+6}{4} \right\rfloor = k + 1.$$

$$\text{Sum} = 2k + (k + 1) + (k + 1) = 4k + 2 = n.$$

- $r = 3$ ($n = 4k + 3$):

$$\left\lfloor \frac{4k+2}{2} \right\rfloor = 2k + 1, \quad \left\lfloor \frac{4k+5}{4} \right\rfloor = k + 1, \quad \left\lfloor \frac{4k+7}{4} \right\rfloor = k + 1.$$

$$\text{Sum} = (2k + 1) + (k + 1) + (k + 1) = 4k + 3 = n.$$

By checking all four residue classes modulo 4 we obtain $S(n) = n$ for every integer n , which proves the claim. □

Exercise 5 — Upper bound sets and Lower bound sets

Let A and B be two nonempty bounded subsets of $\mathbb{R}$. Show that:

1. If $A \subset B$ then $\inf B \leq \inf A$ and $\sup A \leq \sup B$.
2. $\sup(A \cup B) = \max\{\sup A, \sup B\}$, $\inf(A \cup B) = \min\{\inf A, \inf B\}$.

Solution to Exercise 5

1. If $A \subset B$ then $\inf B \leq \inf A$ and $\sup A \leq \sup B$.

Since B is bounded and non-empty, $\sup B$ and $\inf B$ exist, and we have:

$$\forall x \in B : \inf B \leq x \leq \sup B$$

and since $A \subset B$, then

$$\forall x \in A : \inf B \leq x \leq \sup B.$$

Hence, $\sup B$ is an upper bound of A and since $\sup A$ is the least upper bound of A , we get

$$\sup A \leq \sup B$$

On the other hand, we have :

$\inf B$ is a lower bound of A and since $\inf A$ is the greatest lower bound of A , we get

$$\inf B \leq \inf A$$

Therefore we have proved both relations:

$$\sup A \leq \sup B \quad \text{and} \quad \inf B \leq \inf A.$$

□

2. $\sup(A \cup B) = \max\{\sup A, \sup B\}$, $\inf(A \cup B) = \min\{\inf A, \inf B\}$.

- a. Let us show that $A \cup B$ is bounded (this is trivial but will give us useful relations for the next part of the question). Let $x \in A \cup B$, so $x \in A$ or $x \in B$. Thus,

$$\inf A \leq x \leq \sup A \quad \text{or} \quad \inf B \leq x \leq \sup B$$

and therefore

$$\min(\inf A, \inf B) \leq x \leq \max(\sup A, \sup B).$$

Hence $A \cup B$ is bounded, and we clearly have

$$\sup(A \cup B) \leq \max(\sup A, \sup B) \quad \text{and} \quad \inf(A \cup B) \geq \min(\inf A, \inf B).$$

On the other hand, since $A \subset A \cup B$ and $B \subset A \cup B$, we get

$$\sup A \leq \sup(A \cup B) \quad \text{and} \quad \sup B \leq \sup(A \cup B),$$

that is,

$$\sup(A \cup B) \geq \max(\sup A, \sup B).$$

Thus,

$$\boxed{\sup(A \cup B) = \max(\sup A, \sup B)}.$$

□

b. On the other hand, since $A \subset A \cup B$ and $B \subset A \cup B$, we get

$$\inf(A \cup B) \leq \inf A \quad \text{and} \quad \inf(A \cup B) \leq \inf B,$$

because the infimum of $A \cup B$ is a lower bound for $A \cup B$ and thus for each of its subsets. Hence

$$\inf(A \cup B) \leq \min(\inf A, \inf B).$$

Combining this with the opposite inequality obtained above

$$\min(\inf A, \inf B) \leq \inf(A \cup B),$$

we conclude

$$\boxed{\inf(A \cup B) = \min(\inf A, \inf B)}.$$

Therefore we have proved both relations:

$$\sup(A \cup B) = \max(\sup A, \sup B) \quad \text{and} \quad \inf(A \cup B) = \min(\inf A, \inf B).$$

□

Exercise 6 — Using the characterization of $\sup$ and $\inf$

1. Let A and B be two nonempty bounded sets of real numbers. Let

$$C = A + B = \{a + b : (a, b) \in A \times B\}.$$

Show that C is bounded and that

$$\sup C = \sup A + \sup B, \quad \inf C = \inf A + \inf B.$$

2. Assume that A is a nonempty set of non-negative real numbers which is bounded above. Prove that

$$\sup(\sqrt{A}) = \sqrt{\sup A},$$

where $\sqrt{A} = \{\sqrt{x} : x \in A\}$.

Solution to Exercise 6

1. Let A and B be two nonempty bounded sets of real numbers. Let

$$C = A + B = \{a + b : (a, b) \in A \times B\}.$$

Boundedness. Since A and B are bounded, there exist real numbers $m_A \leq M_A$ and $m_B \leq M_B$ such that

$$m_A \leq a \leq M_A \quad \forall a \in A, \quad m_B \leq b \leq M_B \quad \forall b \in B.$$

For any $c = a + b \in C$ we obtain

$$m_A + m_B \leq a + b \leq M_A + M_B,$$

so C is bounded (by $m_A + m_B$ and $M_A + M_B$).

b. Show that

$$\sup C = \sup A + \sup B, \quad \inf C = \inf A + \inf B.$$

Step 1. Upper bound.

For all $a \in A$ and $b \in B$ we have $a \leq S_A$ and $b \leq S_B$. Adding gives $a + b \leq S_A + S_B$. So $S_A + S_B$ is an upper bound of C . Hence

$$\sup C \leq S_A + S_B.$$

Step 2. Reverse inequality using the ε -definition of supremum.

Let $\varepsilon > 0$.

By the definition of supremum:

- Since $S_A = \sup A$, there exists $a \in A$ such that

$$S_A - \varepsilon/2 < a \leq S_A.$$

- Since $S_B = \sup B$, there exists $b \in B$ such that

$$S_B - \varepsilon/2 < b \leq S_B.$$

Add the two inequalities:

$$(S_A - \varepsilon/2) + (S_B - \varepsilon/2) < a + b \leq S_A + S_B.$$

That is,

$$a + b > (S_A + S_B) - \varepsilon.$$

Because $a + b \in C$, we have found an element of C that is within ε of $S_A + S_B$ from below.

This means: For every $\varepsilon > 0$, there exists $c \in C$ such that

$$c > (S_A + S_B) - \varepsilon.$$

Therefore, no number smaller than $S_A + S_B$ can be an upper bound of C .

Hence,

$$\sup C \geq S_A + S_B.$$

Step 3. Conclusion.

From both inequalities we have

$$\sup C = S_A + S_B.$$

Remark (infimum).

The same idea works for infima. Let $m_A = \inf A$ and $m_B = \inf B$. For any $a \in A$, $b \in B$, we have $a \geq m_A$ and $b \geq m_B$, so $a + b \geq m_A + m_B$, giving

$$\inf C \geq m_A + m_B.$$

Using the ε -definition of infimum, for every $\varepsilon > 0$ there exist $a \in A$ and $b \in B$ with

$$a < m_A + \varepsilon/2, \quad b < m_B + \varepsilon/2.$$

Then $a + b < m_A + m_B + \varepsilon$, so no larger number than $m_A + m_B$ can be a lower bound of C , which gives $\inf C \leq m_A + m_B$.

Thus,

$$\boxed{\sup C = S_A + S_B \quad \text{and} \quad \inf C = m_A + m_B.}$$

□

2. Prove that

$$\sup(\sqrt{A}) = \sqrt{\sup A},$$

Step 1. First inequality.

For every $a \in A$, we have $0 \leq a \leq S$. Taking square roots gives $0 \leq \sqrt{a} \leq \sqrt{S}$. So, $\sqrt{S}$ is an upper bound of $\sqrt{A}$. Hence,

$$\sup(\sqrt{A}) \leq \sqrt{S}.$$

Step 2. Second inequality (using epsilon definition of supremum).

(a) **Definition of supremum :**

1. $\forall a \in A, a \leq S$
2. $\forall \epsilon > 0$, there exist $a \in A$, such that $a > S - \epsilon$

We must show that

$$\forall \epsilon > 0, \text{ there exist } \sqrt{a} \in \sqrt{A}, \text{ such that } \sqrt{a} > \sqrt{S} - \epsilon$$

For $0 < \epsilon < \sqrt{S}$.

Compute

$$S - (\sqrt{S} - \epsilon)^2 = S - (S - 2\sqrt{S}\epsilon + \epsilon^2) = 2\sqrt{S}\epsilon - \epsilon^2.$$

Because $0 < \epsilon < \sqrt{S}$ this quantity is positive. By the definition of supremum (applied to $S = \sup A$), for the positive number

$$S - (\sqrt{S} - \epsilon)^2 > 0$$

there exists some $a \in A$ with

$$a > (\sqrt{S} - \epsilon)^2.$$

Taking nonnegative square roots of both sides yields

$$\sqrt{a} > \sqrt{S} - \epsilon.$$

Since this holds for every $\epsilon > 0$, it follows that no number smaller than $\sqrt{S}$ can be an upper bound of $\sqrt{A}$; hence

$$\sup(\sqrt{A}) \geq \sqrt{S}.$$

Combining the two inequalities gives

$$\sup(\sqrt{A}) = \sqrt{S} = \sqrt{\sup A},$$

as required.

□

Exercise 7 — Infimum and Supremum

Find, if they exist, the infimum and supremum of each of the following sets:

$$\begin{aligned} A &= \{x \in \mathbb{N} \mid x^2 - x - 6 > 0\}, & B &= \{n \cos(n\pi); n \in \mathbb{N}\}, \\ C &= \{x \in \mathbb{Q} \mid x^3 \leq 8\}, & D &= \left\{ \frac{(-1)^n}{n} e^{-n}; n \in \mathbb{N}^* \right\}, \\ E &= \left\{ \frac{n}{mn+1}; (m, n) \in (\mathbb{N}^*)^2 \right\}, & F &= \{\sqrt{n} - \lfloor \sqrt{n} \rfloor; n \in \mathbb{N}\}, \\ G &= \left\{ \frac{m}{n} + \frac{4n}{m}; m, n \in \mathbb{N}^* \right\}, & H &= \{2^x + 2^{1/x}; x > 0\}. \end{aligned}$$

Solution to Exercise 7

1. For the set A , it is easy to verify that if $x \in \mathbb{R}$, then

$$x^2 - x - 6 > 0 \iff x \in (-\infty, -2) \cup (3, +\infty).$$

But in our case $x \in \mathbb{N}$. Hence

$$A = \{4, 5, 6, \dots\} = \mathbb{N} \setminus \{0, 1, 2, 3\}.$$

Therefore,

$$\sup A = +\infty \quad \text{and} \quad \inf A = \min A = 4.$$

That's mean $\sup A$ is not exist.

2. We note that

$$B = \{n \cos(n\pi) : n \in \mathbb{N}\} = \{n(-1)^n : n \in \mathbb{N}\}$$

Thus, B is neither bounded above nor below, and hence

$$\sup B = +\infty \quad \text{and} \quad \inf B = -\infty.$$

That's mean $\sup B$ and $\inf B$ are not exists.

3. $C = \{x \in \mathbb{Q} \mid x^3 \leq 8\}$ We have $x^3 \leq 8$ with $x \in \mathbb{Q}$.

Then

$$C = \{x \in \mathbb{Q} \mid x \leq 2\}.$$

2 is an upper bound of C and $2 \in C$.

As a result

$$\sup C = 2$$

Since the set C is not bounded below $\inf C$ not exist.

4. For the set D .

Notice that

$$0 < \frac{1}{n} \leq 1 \quad \text{and} \quad 0 < e^{-n} \leq 1 \quad \forall n \in \mathbb{N}^*,$$

so

$$-1 \leq \frac{(-1)^n}{n} \leq 1$$

and therefore

$$-1 \leq \frac{(-1)^n e^{-n}}{n} \leq 1.$$

Hence

$$-1 \leq x \leq 1 \quad \forall x \in D,$$

so D is bounded and $\sup D$ and $\inf D$ exist.

Goal: find $\sup D$ and $\inf D$.

We write

$$D = D_1 \cup D_2$$

where

$$D_1 = \left\{ U_n = \frac{e^{-2n}}{2^n} : n \in \mathbb{N}^* \right\}, \quad D_2 = \left\{ V_n = \frac{-e^{-2n-1}}{2n+1} : n \in \mathbb{N} \right\}.$$

Step A — Analysis of D_1 .

Compute the ratio of successive terms of D_1 :

$$\frac{U_{n+1}}{U_n} = \frac{e^{-2(n+1)}/2^{n+1}}{e^{-2n}/2^n} = \frac{e^{-2}}{2}.$$

Since

$$\frac{e^{-2}}{2} = \frac{1}{2e^2} < 1,$$

the sequence (U_n) is strictly decreasing. Also $U_n > 0$ for all n . Hence the largest element of D_1 is the first term U_1 , and the terms decrease to 0. Therefore

$$\sup D_1 = U_1 = \frac{e^{-2}}{2}, \quad \inf D_1 = 0.$$

Step B — Analysis of D_2 .

All terms V_n are negative. Consider the absolute values:

$$|V_n| = \frac{e^{-(2n+1)}}{2n+1}.$$

Compare successive absolute values:

$$\frac{|V_{n+1}|}{|V_n|} = \frac{e^{-(2n+3)}/(2n+3)}{e^{-(2n+1)}/(2n+1)} = e^{-2} \cdot \frac{2n+1}{2n+3}.$$

Since $\frac{2n+1}{2n+3} < 1$ and $e^{-2} < 1$, we get

$$\frac{|V_{n+1}|}{|V_n|} < 1,$$

so $|V_n|$ is strictly decreasing. Consequently the negative numbers V_n are strictly increasing (become less negative) as n increases. The smallest (most negative) value appears at the smallest index. Because D_2 was defined with $n \in \mathbb{N}$ (including 0), the smallest index is $n = 0$. Thus

$$\min D_2 = V_0 = -\frac{e^{-1}}{1} = -e^{-1}.$$

As $n \rightarrow \infty$, $V_n \rightarrow 0$ from below, so 0 is an upper bound of D_2 but is not attained. Therefore

$$\inf D_2 = V_0 = -e^{-1}, \quad \sup D_2 = 0.$$

Step C — Combine results for D .

We have

$$\sup D_1 = \frac{e^{-2}}{2}, \quad \inf D_1 = 0,$$

and

$$\sup D_2 = 0, \quad \inf D_2 = -e^{-1}.$$

Since $D = D_1 \cup D_2$, the supremum of D is the larger of $\sup D_1$ and $\sup D_2$. Because $\frac{e^{-2}}{2} > 0$, we get

$$\sup D = \sup D_1 = \frac{e^{-2}}{2}.$$

The infimum of D is the smaller of $\inf D_1$ and $\inf D_2$. Since $\inf D_1 = 0$ and $\inf D_2 = -e^{-1} < 0$, we obtain

$$\inf D = \inf D_2 = -e^{-1}.$$

Answer.

$$\sup D = \frac{e^{-2}}{2}, \quad \inf D = -e^{-1}.$$

□

5. For the set E

$$E = \left\{ \frac{n}{mn+1} : m, n \in \mathbb{N}^* \right\}.$$

where $\mathbb{N}^* = \{1, 2, 3, \dots\}$.

1. Basic bounds. For any fixed positive integers m, n we have $mn+1 > mn$, so

$$0 < \frac{n}{mn+1} < \frac{n}{mn} = \frac{1}{m} \leq 1.$$

Therefore every element of E lies in $(0, 1)$. In particular 0 is a lower bound and 1 is an upper bound of E . Hence $\inf E$ and $\sup E$ exist and satisfy

$$0 \leq \inf E \leq \sup E \leq 1.$$

2. Find the supremum. For a fixed n , the map $m \mapsto \frac{n}{mn+1}$ is strictly decreasing in m (because the denominator increases), so the largest value for that fixed n occurs at $m = 1$:

$$\max_{m \geq 1} \frac{n}{mn+1} = \frac{n}{n+1}.$$

Thus $\frac{n}{n+1} \in E$ for every n . Note

$$\frac{n}{n+1} = 1 - \frac{1}{n+1} \longrightarrow 1 \quad (n \rightarrow \infty),$$

so we can make elements of E arbitrarily close to 1 from below.

Formal ε -argument: let $\varepsilon > 0$. Choose n with $1/(n+1) < \varepsilon$ (possible since $1/(n+1) \rightarrow 0$). Then

$$\frac{n}{n+1} = 1 - \frac{1}{n+1} > 1 - \varepsilon,$$

and $\frac{n}{n+1} \in E$. Therefore no number smaller than 1 can be an upper bound of E . Combining this with the fact that 1 is an upper bound, we get

$$\boxed{\sup E = 1.}$$

Note $1 \notin E$, so the supremum is not attained (there is no maximum).

3. Find the infimum. All elements of E are positive, so 0 is a lower bound. To show 0 is the greatest lower bound, show we can find elements of E arbitrarily small. Take $n = 1$. Then

$$\frac{1}{m \cdot 1 + 1} = \frac{1}{m+1} \in E$$

and $\frac{1}{m+1} \rightarrow 0$ as $m \rightarrow \infty$. More precisely, for any $\varepsilon > 0$ choose m with $m+1 > 1/\varepsilon$ (possible since m can be arbitrarily large). Then

$$\frac{1}{m+1} < \varepsilon,$$

so E contains an element $< \varepsilon$. Hence no positive number can be a lower bound smaller than or equal to all elements of E . Therefore

$$\boxed{\inf E = 0.}$$

Again $0 \notin E$, so the infimum is not attained (there is no minimum).

Final answer :

$$\inf E = 0, \quad \sup E = 1,$$

and neither 0 nor 1 belongs to E . □

6. For the set F

$$F = \{\sqrt{n} - \lfloor \sqrt{n} \rfloor; n \in \mathbb{N}\}.$$

Step 1. Recall the definition of the integer part.

For any real number $x \geq 0$,

$$\lfloor x \rfloor \text{ is the unique integer such that } \lfloor x \rfloor \leq x < \lfloor x \rfloor + 1.$$

Hence for all $x \geq 0$,

$$0 \leq x - \lfloor x \rfloor < 1.$$

We apply this to $x = \sqrt{n}$.

Step 2. Bounding the elements of F .

For every $n \in \mathbb{N}$,

$$0 \leq \sqrt{n} - \lfloor \sqrt{n} \rfloor < 1.$$

Therefore all elements of F lie in the interval $[0, 1)$. So 0 is a lower bound and 1 is an upper bound of F . Thus

$$0 \leq \inf F \leq \sup F \leq 1.$$

Step 3. Determine $\inf F$.

Take $n = k^2$ for $k \in \mathbb{N}$. Then $\sqrt{n} = k$, so

$$\sqrt{n} - \lfloor \sqrt{n} \rfloor = k - k = 0.$$

Hence $0 \in F$. Since all elements are nonnegative, 0 is the smallest element. Therefore

$$\boxed{\inf F = 0.}$$

Step 4. Determine $\sup F$.

For each integer $k \geq 0$, consider all n such that $\lfloor \sqrt{n} \rfloor = k$. This means

$$k \leq \sqrt{n} < k + 1,$$

so

$$k^2 \leq n < (k + 1)^2.$$

For these n , we have

$$\sqrt{n} - \lfloor \sqrt{n} \rfloor = \sqrt{n} - k.$$

The smallest value occurs when $n = k^2$ (which gives 0), and the largest value occurs when n is just before $(k + 1)^2$.

Thus for each k , the values of $\sqrt{n} - \lfloor \sqrt{n} \rfloor$ fill an interval $[0, 1)$ but never reach 1.

Therefore, for any $\varepsilon > 0$, we can find n such that

$$\sqrt{n} - \lfloor \sqrt{n} \rfloor > 1 - \varepsilon,$$

but the value 1 itself never occurs. Hence no number less than 1 can be an upper bound, and 1 is the least upper bound.

So

$$\boxed{\sup F = 1.}$$

Note that $1 \notin F$, since $\sqrt{n}$ is never exactly one unit larger than an integer.

Final answer:

$$\inf F = 0, \quad \sup F = 1.$$

and $F \subset [0, 1)$. □

7. For the set G

$$G = \left\{ \frac{m}{n} + \frac{4n}{m}; m, n \in \mathbb{N}^* \right\}.$$

To find $\sup G$ and $\inf G$, let us set

$$a = \frac{m}{n} > 0.$$

Then each element of G can be written as

$$\frac{m}{n} + \frac{4n}{m} = a + \frac{4}{a}.$$

So we define the function

$$f(a) = a + \frac{4}{a}, \quad a > 0.$$

To find the minimum value of f , we compute the derivative:

$$f'(a) = 1 - \frac{4}{a^2}.$$

We set $f'(a) = 0$ to find critical points:

$$1 - \frac{4}{a^2} = 0 \quad \Rightarrow \quad a^2 = 4 \quad \Rightarrow \quad a = 2.$$

We evaluate f at $a = 2$:

$$f(2) = 2 + \frac{4}{2} = 4.$$

For $a \rightarrow 0^+$, we have $f(a) \rightarrow +\infty$, and for $a \rightarrow +\infty$, we have $f(a) \rightarrow +\infty$.

Therefore, the function f reaches its minimum value 4 at $a = 2$, and it is unbounded above.

$\inf G = 4, \quad \sup G \text{ is not exist.}$
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□

8. For the set

$$H = \{2^x + 2^{1/x}; x > 0\}$$

Use the AM–GM inequality with $a = 2^x > 0$ and $b = 2^{1/x} > 0$:

$$\frac{a+b}{2} \geq \sqrt{ab} \quad \Rightarrow \quad a+b \geq 2\sqrt{ab}.$$

Here $ab = 2^{x+\frac{1}{x}}$, so

$$2\sqrt{ab} = 2 \cdot 2^{\frac{x+\frac{1}{x}}{2}} = 2^{1+\frac{x+\frac{1}{x}}{2}}.$$

Therefore

$$2^x + 2^{1/x} \geq 2 \cdot 2^{\frac{x+\frac{1}{x}}{2}}.$$

Now use the elementary inequality $x + \frac{1}{x} \geq 2$ for $x > 0$, with equality only at $x = 1$. Hence

$$2^{\frac{x+\frac{1}{x}}{2}} \geq 2^{\frac{2}{2}} = 2,$$

and combining with the previous line gives

$$2^x + 2^{1/x} \geq 2 \cdot 2 = 4.$$

Equality holds at $x = 1$ (since then $2^x = 2^{1/x} = 2$), so $4 \in H$ and is the smallest value.

Finally, as $x \rightarrow \infty$ we have $2^x \rightarrow \infty$ while $2^{1/x} \rightarrow 1$, and as $x \rightarrow 0^+$ we have $2^x \rightarrow 1$ while $2^{1/x} \rightarrow \infty$. Thus elements of H can be made arbitrarily large, so H is unbounded above.

$\inf H = 4, \quad \sup H \text{ is not exist.}$
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□